

Element B

Documentation and Analysis Of Prior Solution Attempts

Problem Statement

Reports indicate that 42,000 American homeowners annually face recurring neck, shoulder, and back injuries from picking up and disposing of leaves in their yards¹. Disregarding this task causes leaves to accumulate in yards, smothering the grass and blocking the sunlight needed for proper growth. Overall, leaf removal is a task that can cause physical strain when done and, if neglected, can affect the yard's health, leading to grass death in the fall.

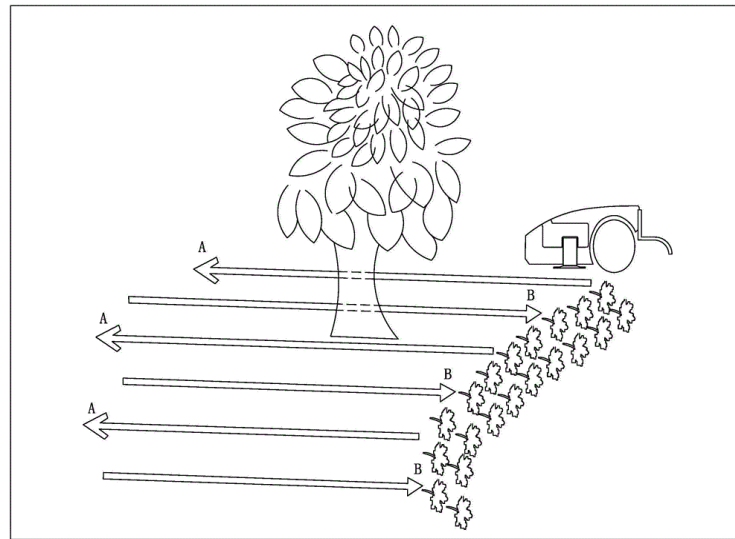
Introduction: Several companies have attempted to address the problem described above by developing and testing various solutions. This report examines those existing approaches, highlighting their shortcomings and identifying aspects in which they have been effective in solving the problem.

Patents:

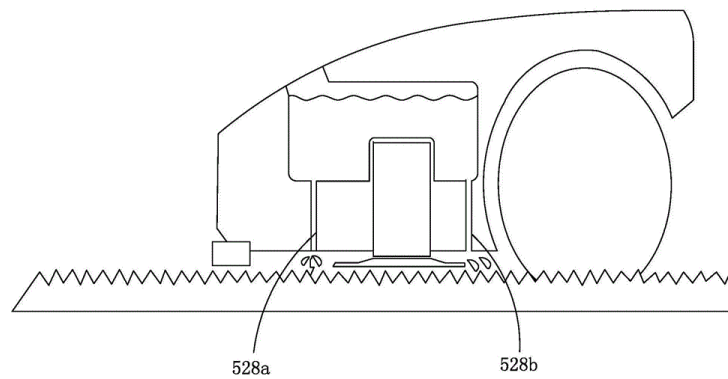
Some products, although not yet sold or manufactured, still offer ways to solve the leaf removal problem. This section analyzes existing patents to determine which ideas already exist and where they are lacking.

¹ Don't let raking leaves become a pain in the neck, back or spine. Luminis Health. (n.d.). <https://www.luminishealth.org/en/blog/dont-let-raking-leaves-become-a-pain-in-the-neck-back-or-spine?>

1. Self-Gardening Robot



100



100

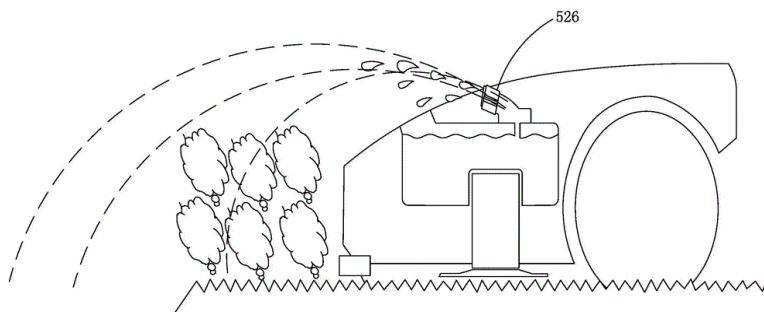


Figure B.1: Automated self-gardening robot²

² Zhang, X. (2019). 自移动园艺机器人及其系统 [Self-moving horticultural robot and system] (China Patent No. CN108541308B). China National Intellectual Property Administration.
<https://patents.google.com/patent/CN108541308B/en>

The focus of the patent displayed in Figure B.1 is on a self-gardening robot capable of performing many everyday yard tasks. The robot has a charging station and is able to map out the area required for cleanup. It also has the option to place signals at various parts of the yard so the robot knows where to stay for an extended period of time for cleanup.

The robot is multifunctional, coming with other add-on accessories that can be used for other yard work.

Patent Number: CN108541308B

Pro's of patent:

- Can pick up leaves.
- Can do multiple yard tasks(not at the same time).
- Can set up boundary signals, so it adapts to different yard types.
- If needed, the owner can move the robot with a device.

Con's of patent:

- Requires the purchase of extra parts for specific tasks.
- The user has to dispose of the leaves themselves once the robot reaches storage capacity.
- Has to be moving in a specific position in order to pick up the leaves.

Patent Critique:

This patent shows a very functional design, but it lacks a holistic explanation as to how the components work together. It describes a lot of its components and their contributions to the machine's work, but it does not fully explain how those parts work together.

Overall, the design works very well to solve the problem, requiring no physical labor by the user, minimizing injury during the cleanup process. However, the user will have to clear out the leaves from the machine by themselves, creating the potential for injury.

2. Autonomous All-Purpose Machine

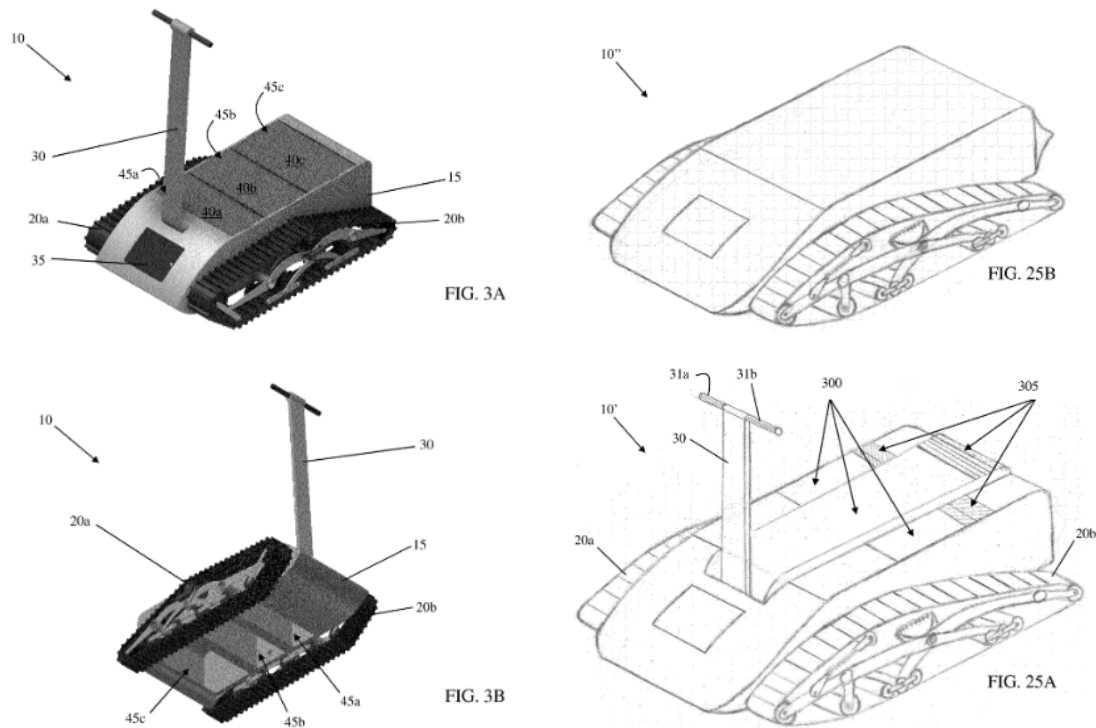


Figure B.3: Isometric view of the all-purpose autonomous and remote control machine³

The patent shown in Figure B.3 shows an all-purpose machine designed to perform various tasks, such as leaf-vacuum shredding, mowing, and snow removal. The machine does this by using multiple attachment modules, which have to be purchased additionally. The machine can operate autonomously or be controlled remotely by the user's device.

Patent Number: US20140121881A1

Pro's of patent:

- Reduces physical effort
- Saves time on manual leaf removal
- Flexible through multiple yard tasks

Con's of patent:

- General-purpose design may need adaptations for optimal leaf removal
- Potentially high cost
- Requires additional attachments for other tasks

³ Diaz del Castillo, D. A. (2014). Autonomous and remote control all purpose machine. U.S. Patent Application No. US20140121881A1. <https://patents.google.com/patent/US20230154321A1/en>

Patent Critique:

The design benefits the user by reducing physical strain because tasks can be completed autonomously or through remote control rather than manual labor.

Overall, the machine improves user safety and comfort by limiting direct physical effort. However, the need for additional attachments may increase cost and setup time, which could affect overall convenience for the user.

3. Suction and Blowing Device

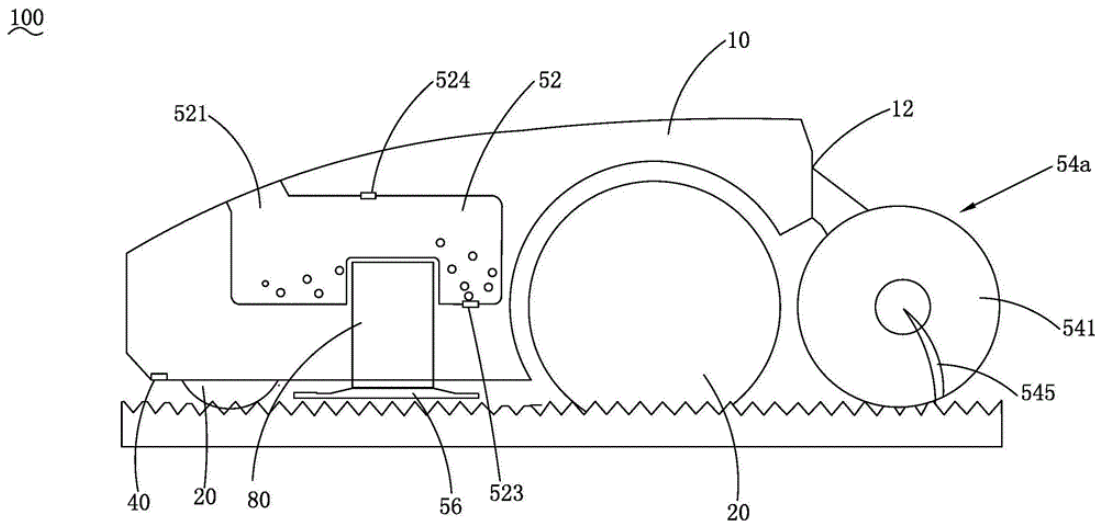


Figure B.2: Suction and Blowing Device⁴

The suction blowing device follows the principles of a vacuum cleaner but adds features such as a larger containment unit for leaves and other debris. The vacuum, which features a central unit for blowing and sucking, can be equipped with various attachments. One part can be used to catch smaller particles of dust. To have a more substantial impact, it has two separate fans to handle the sucking and blowing module of the vacuum.

Pro's of patent:

- Can add modifications for specific needs
- Ability to suck up and store leaves
- Stronger fan power

Con's of patent:

- With larger fans, the tubes have to become larger, increasing the size of the entire unit

⁴ CN206538723U - Suction and blowing device - Google Patents. (2016, April 6). Google.com. [https://patents.google.com/patent/CN206538723U/en?q=\(leaf+removal\)&oq=leaf+removal&page=1](https://patents.google.com/patent/CN206538723U/en?q=(leaf+removal)&oq=leaf+removal&page=1)

- With so much pressure, blockage in the tubes can often happen

Patent Critique:

Overall, the design is effective for collecting leaves and debris and reduces the physical effort required from the user. However, the use of multiple fans and attachments may increase mechanical complexity, and handling or maintaining these components could create potential safety or usability concerns for the user. The complexity also poses many points of failure for the machine, meaning that it could require frequent maintenance, which could be a financial and physical burden.

4: Autonomous All-Purpose Machine, Second Model

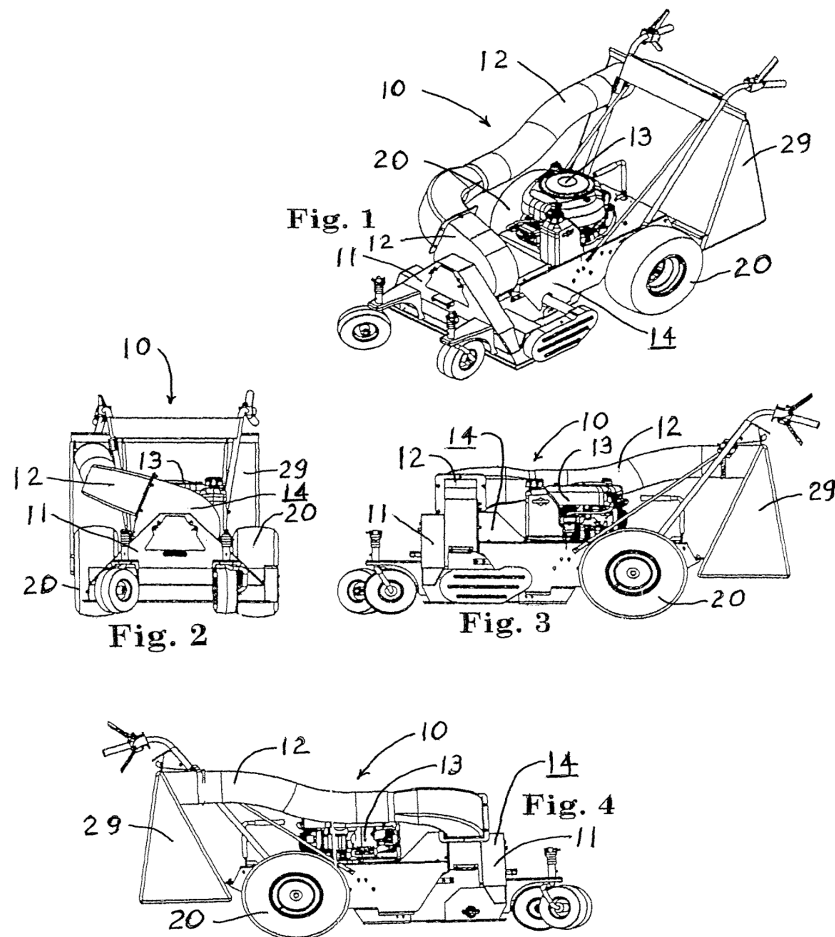


Figure B.4: Diagram of the model of a second all-purpose machine⁵

⁵ Diaz del Castillo, D. A. (2014). Autonomous and remote control all purpose machine. U.S. Patent Application No. US20140121881A1. <https://patents.google.com/patent/US20230154321A1/en>

Figure B.4 describes a lawn debris vacuum machine that uses suction to draw leaves and other yard debris into a chute and through to a collection. It includes a thrasher to break up larger debris that might clog the chute. The machine also consists of a rotating brush at the chute mouth to draw debris into the intake more efficiently.

Patent Number: US20140121881A1

Pro's of patent:

- Efficient debris collection
- Shreds leaves to reduce volume for disposal
- Not limited to only leaves

Con's of patent:

- Complex design
- Higher cost
- May struggle with obstacles

Patent Critique:

This machine reduces physical strain on the user by not requiring manual lifting or raking. Features such as the thrasher help move and break down debris efficiently, lowering the risk of clogs and reducing the need for user intervention.

Overall, the design improves user safety by minimizing lifting and bending. However, clearing the collection unit could still be a potential risk of injury.

5: Leaf Collector Robot

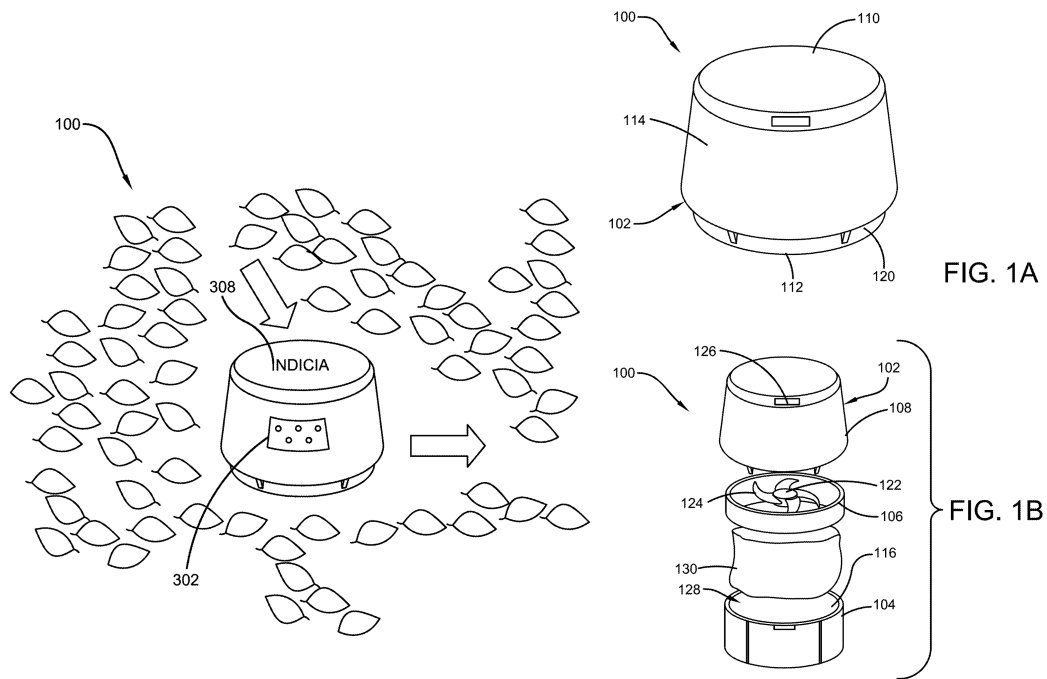


Figure B.5: Diagram of leaf robot⁶

Figure B.5 shows the design for a lawn maintenance system that includes an autonomous mower with a docking station. The docking station can spray water onto the mower to clean it after docking. The mower would charge and be cleaned at a single station, and this patent adds to an existing autonomous mower. This mower also has another theorized function of cleaning up debris on lawns, which is the solution needed for the problem needed to solve.

Patent Number: US20240081201A1

Pro's of patent:

- Has self-cleaning capabilities
- Uses electricity rather than manpower or gas
- Guides itself to the docking station

Con's of patent:

- Needs to be set by a water source (such as a spout)
- Needs to be set by a power outlet
- A small device that may require frequent clearing

⁶ Techtronic Cordless GP. (2024). Lawn maintenance system having an autonomous lawn mower and a docking station (U.S. Patent Application No. US20240081201A1). U.S. Patent and Trademark Office.
<https://patents.google.com/patent/US20240081201A1>

Patent Critique:

The design benefits the user by reducing physical strain by operating independently and eliminating the need for manual labor.

Overall, the system improves user safety by minimizing direct interaction with moving parts and debris. However, the requirement for access to both a power outlet and a water source limits placement flexibility, and the mower's small size may require frequent clearing, which could still expose the user to minor physical strain or injury.

6: Leaf Collector

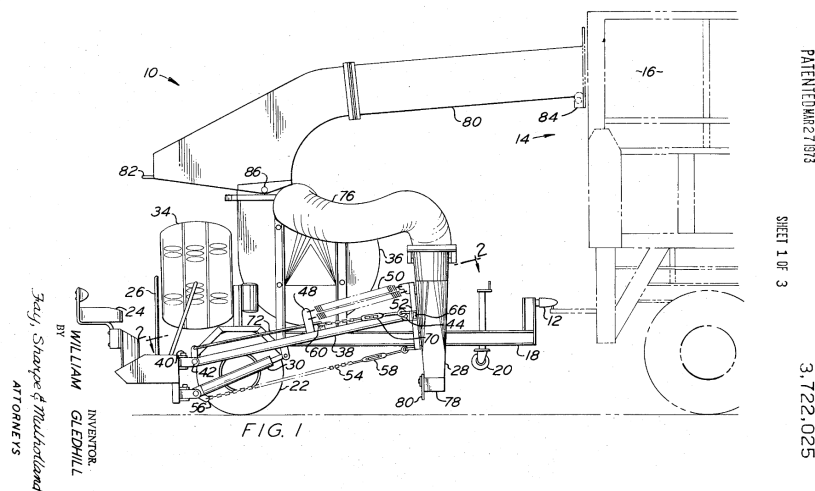


Figure B.5: Design of leaf collector⁷

Figure B.5 is for a heavy-duty leaf-collecting device intended for mounting on a large vehicle, such as a truck. This solves the problem of leaves on the road, which is a significant issue many people in neighborhoods face. This design uses a high-powered vacuum that will collect debris into a collection container. One prominent feature of this design is the nozzle collecting into the chamber using a vacuum, which can be added to the design for a collection and intake.

Pros of patent:

- Works on a spring mechanism, allowing the device to work efficiently with constant operation when needed
- Works for a larger volume of debris and leaves

⁷ Gledhill, W. (1973). Leaf collector (U.S. Patent No. US3722025A). U.S. Patent and Trademark Office. <https://patents.google.com/patent/US3722025A/en>

Cons of patent:

- Equipment durability is a risk when dealing with roadside leaves
- To keep the device in check, a nozzle has to be used and checked manually to keep the device running, leading to causes of intensive care and human error
- Very complex

Patent Critique:

The machine handles high volumes of debris and reduces physical work for operators, but it's too big and complex for personal use. Users still need to check the nozzle and maintain it, which can cause errors. It's good for street and commercial use, but not for home use.

Current Solutions:

Some products are currently available to address leaf removal and can be purchased. This section examines prior solutions currently available in the market similar to this problem.



Figure B.6: Scott Lawn Sweeper's Lawn Sweeper⁸

B.6 shows a machine that can pick up leaves and other yard waste such as pine cones and small branches. It has the strength of picking up items rather than just leaves, but it still requires effort to push the sweeper all over the yard.

Product Price: \$179.99.

Where to buy:

- Home Depot
- Ace Hardware
- Amazon

⁸ Ace Hardware. (n.d.). Scotts Lawn Sweeper (Item # 7821416). Retrieved December 9, 2025, from <https://www.acehardware.com/departments/lawn-and-garden/lawn-mowers/graself-removalss-catchers/7821416?>

Brand: Scott's

Pros:

- Picks up debris other than leaves
- Can collect larger items

Cons:

- Requires self-removal of the bag, which can be heavy.
- Still needs physical labor by the user.



Figure B.7: Dr. Power Leaf Vacuum⁹

The Dr. Power Leaf Vacuum (Figure B.7) has much more power and eliminates most of the manual labor required for leaf removal. It has one of the largest storage capacities among other brands, as well as a larger nozzle for collecting leaves. Inside the nozzle is a fan not only pulling the leaves in but shredding them as they go into the bag. When the leaves are cut into smaller pieces, there is more room to fit more leaves, making leaf removal quicker.

Price: \$1,599.99

Where to buy:

- Home Depot
- DR. Power Equipment

Brand: Dr. Power

Pros:

⁹ DR Power Equipment. (n.d.). DR Leaf and Lawn Vacuum – Walk-Behind (WL33206DMN). Retrieved December 9, 2025, from <https://www.drpower.com/Power-Equipment/Leaf-Blowers-%26-Vacuums>

- Makes removal quicker by a larger volume of holding
- Shredded leaves
- Largest capacity among competitors

Cons:

- Very expensive
- Manually removing the leaf storage(causes physical labor)



Figure B.8: ZEGJAW Artificial Turf Sweeper¹⁰

This machine is an electric alternative to a lawn sweeper, offering effortless movement and requiring less user force. This product picks up leaves and debris as the user talks the machine around the lawn, using an electric motor for suction.

Price: \$229.99

Where to buy:

- Amazon
- Ace Hardware

Brand: ZEGJAW

Pros:

- Less physically demanding than manual pickup
- Smooth movement
- Lower Price point

Cons:

- Smaller storage capacity
- Still physical labor to move the machine around

¹⁰ Amazon. (n.d.). ZEGJAW Artificial Grass Sweeper, 5-position adjustment [Product page]. Amazon. <https://www.amazon.com/ZEGJAW-Artificial-5-Position-Adjustment-Collection>



Figure B.9: BLACK+DECKER Leaf Blower & Vacuum¹¹

This machine is a vacuum and a blower for leaves as a single product. The large tube is attached to a storage unit that can be put on as a backpack. Even as a backpack, it still has a large storage compartment.

Price: \$99

Where to buy:

- Ace Hardware
- Home Depot
- Walmart
- BLACK+DECKER
- Amazon

Brand: BLACK+DECKER

Pros:

- Low Price
- Very versatile
- Big storage

Cons:

- Having a backpack can cause back pain
- Heavy storage to dispose of for the user

¹¹ Amazon. (n.d.). BLACK+DECKER BEBL7000 Blower/Mulcher [Product page]. Amazon.
https://www.amazon.com/BLACK-DECKER-BEBL7000-Blower-Mulcher/dp/B07253C39S/ref=sr_1_10?

Solution Matrix

Similar Solution Matrix		Specifications	Price Range	Large storage	Amount of Effort to be put in	Accessibility	Yard Flexibility	Needs an energy source	Durability	Ergonomics	Adjustability	Safety	Uniqueness	ROI (Worth the money?)	Compactness	Design Appeal	Ease of Repair	Noise & Emissions	Manufacturing Efficiency
Sources	Solutions & Designs																		
	Solution A																		
Ace Hardware Scotts																			
Dr. Power Electric Pu																			
ZEGJAW Artificial Tu																			
BLACK+DECKER Le																			
Self Gardening Robo																			
Suction Blowing Devi																			
All Purpose Machine																			
All Purpose Machine,																			
Leaf Collector Robot																			
Leaf Collector																			

Figure B.10: Similar Solution Matrix

There are many different solutions/prototypes for leaf removal and each design has its own advantages and disadvantages. A similar solution matrix (Figure B.10) was created to evaluate them.

The matrix compared the patented designs and commercially available products across 17 different categories that describe the product. The matrix uses a red, yellow, and green system to describe performance levels, with red being the lowest and green being the highest. This makes it very easy to see which solutions excel in different areas and allows for quick comparisons.

The Black+Decker leaf vacuum performed best, earning 13 greens and four reds, while the leaf collector robot performed worst, earning 10 reds and only seven greens. The matrix shows the inefficiencies and technological advancements in today's leaf removal technology.

Conclusion

All of the solutions discussed in this report make it easier for the user to pick up leaves and dispose of them. The patents focused more on an automation approach, such as the Self-Gardening Robot (figure B.1), letting a machine do the work without human input. Current solutions, such as Lawn Sweeper (figure B.6), still require human effort, but require less work to remove leaves.

The solution matrix provides what is needed to be in a new solution, but also shows what many of the solutions lack, such as the ease of repair in multiple machines like the one in Figure B.10. Overall, this research will help take a step closer to creating a new, unique solution to leaf removal.